

Report showses the **modifications made during** **the implementation stages**

Models and arrangement of project files:

- I created a model for perfume and split it into two parts: the glass part and plastic part in which the glass part is a complete bottle but the lower plastic part is created by split by slice plane and .
- I created two packages in different shapes and animated them according to their shapes.
- I created a curtain and animated it .
- I created a petal model and used it in the first scene but to make render time

fast , I used petal texture on the plane and depend on opacity to hide the background of petals.

- All scenes were in a single 3d max file but after advice from our trainer ,I separated four scenes into four files resulting in work becoming easier especially when working with light and render.

Lighting and background:

- Used Arnold light type :
 - quad (plane light) to make studio lighting consist of key light , fill light and rim light and I added light above models in the fourth scene .
 - Photometric type and upload a file of light to it in the first scene.
- Background is plane then converted into editable poly and work on edge and make

chamfer for middle edge and increase segment to make it curvature and smoothing it.

Animation:

- Keyframes by work directly on timeline were applied in all four scenes, in addition to animation path constraint for animating petals around perfume in scene 2 and using helix in spline as path for this animation. Added cloth modifier and used simulate and create keys to add this simulation as keyframes on timeline for curtain in first scene .
- Used PF source in particle system to make petals fall down from top to down and POmniFlect in deflectors in space wraps to make petals stand on ground when they fall down in the fourth scene.
- Total keyframes for four scenes: 1299 frames.
- Time configuration setting for animation:

- Frame rate is 30FPS .
- NTSC

Camera:

- Used physical camera in standard camera.
- Used four cameras where each scene contains one physical camera.

Renderer : is Arnold

- Used ActiveShade Mode for testing material and lighting.
- Used Production Rendering Mode and resolution is 1902*1080 for Render output.

Useful modifier solved challenges:

- Noise modifier for curtain to add detail and complexity to 3D models.
- Turbosmooth modifier to make models more smooth and increase polygons to eliminate sharp edges and create a more polished look for models.
- Arnold properties :
There were some challenges as make model effect by certain light and exclude another light effect on same scene and in same place by apply this modifier on model and include and exclude effect of light or shadow solve these challenges.
- Edit poly :
Make the object more controlled by vertex ,edge , polygon and element at sub-object level .

Post Production :

- Used blender as video editor to convert images to video.
- Video duration : 1 minute and 15 seconds.